Kubernetes Architecture Explained Simply

A beginner-friendly guide to how Kubernetes (K8s) is organized and how its parts work together.

1. What is Kubernetes?

Kubernetes (often written **K8s**) is a system that automatically runs, manages, scales and repairs your applications. Your applications are packaged in **containers**, and Kubernetes spreads them across many computers so you don't have to manage each one by hand.

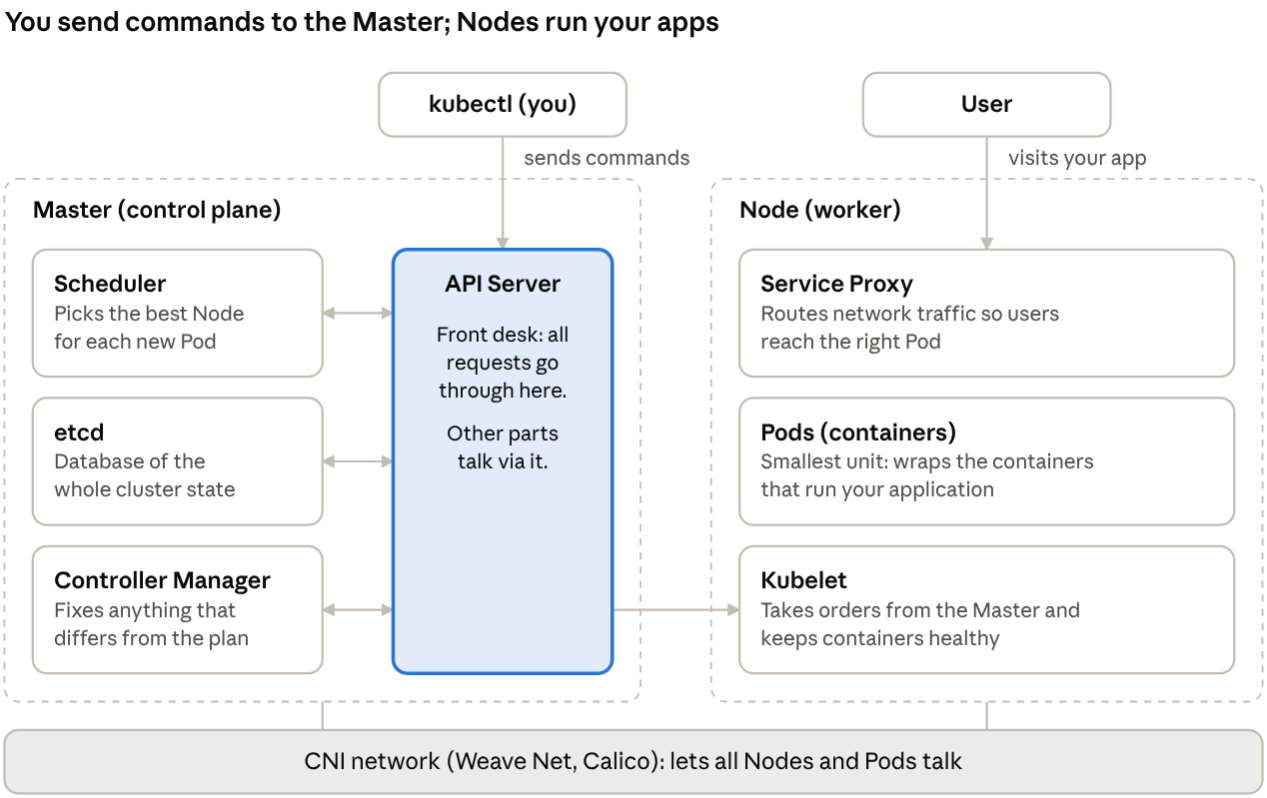
Restaurant analogy: The **Master** is the management office, the **Nodes** are the kitchens where the real cooking happens, the **containers** are the dishes being prepared, and the **User** is the customer.

2. The Big Picture

The cluster has two main sides:

- **Master (Control Plane):** the brain that decides what should run and where.
- **Nodes (Worker Machines):** the servers that actually run your applications.

You, the administrator, talk to the cluster using **kubectl**. End users reach your running apps through the Node's **Service Proxy**. All machines talk to each other over the **CNI network**.



Kubernetes architecture · Master, Node, user and CNI network

You send commands with kubectl to the API Server, which stores state in etcd; the Scheduler and Controller Manager react, and the Kubelet on a Node runs the Pods that users reach through the Service Proxy.

3. Master Components (The Brain)

API Server

The **front desk** of the cluster. Every request, from kubectl or from other components, goes through it. It is the only component that talks to etcd directly, and all other parts communicate via the API Server.

etcd

The cluster's **memory or notebook**. It is a database that stores the entire state of the cluster: what is running, what should be running, and all configuration.

Scheduler

The **planner**. When a new application needs to run, the Scheduler looks at the available Nodes and picks the one with enough CPU, memory and other resources.

Controller Manager

The **supervisor**. It continuously compares the desired state with the actual state. For example, if you asked for 3 copies of your app and one crashes, it creates a replacement automatically.

4. kubectl (Your Remote Control)

`kubectl` is the command-line tool you use to give instructions, such as "run this app", "scale to 5 copies" or "show me what's running". It sends these commands to the API Server.

Example commands:

- `kubectl get pods` shows running pods
- `kubectl apply -f app.yaml` deploys an application
- `kubectl scale deployment myapp --replicas=5` runs 5 copies

5. Node Components (The Workers)

Kubelet

The **Node's manager**. It listens to instructions from the Master (via the API Server), starts the containers, and reports their health back.

Pods and Containers (the orange boxes)

A **Pod** is the smallest unit in Kubernetes. It wraps one or more containers that run your actual application.

Service Proxy (kube-proxy)

The **traffic director**. It manages network rules so that requests from users (or other apps) reach the correct Pod, even when Pods are moved or replaced.

6. User

The end user who visits your application. Their request arrives at the Node, passes through the Service Proxy, and is delivered to a running container.

7. CNI Network (Bottom Arrow)

CNI stands for **Container Network Interface**. Plugins such as **Weave Net** and **Calico** create the network that lets all Nodes and Pods communicate with each other across the cluster.

8. Step-by-Step: How It All Works Together

1. You run a command with **kubectl** (for example, deploy my app).
2. The **API Server** receives the request and saves the desired state in **etcd**.
3. The **Scheduler** notices a new Pod without a Node and chooses the best Node for it.
4. The **Kubelet** on that Node starts the container.
5. The **Controller Manager** keeps watching and fixes anything that breaks, such as restarting crashed Pods.
6. The **User** reaches the application through the **Service Proxy**.

9. Quick Reference Table

Component	Where	Role	Analogy
API Server	Master	Entry point for all requests	Front desk
etcd	Master	Stores cluster state	Notebook
Scheduler	Master	Chooses a Node for new Pods	Planner
Controller Manager	Master	Keeps actual state equal to desired state	Supervisor
kubectl	Your computer	Sends commands	Remote control
Kubelet	Node	Runs and monitors containers	Node manager
Pod	Node	Runs your application	The dish
Service Proxy	Node	Routes network traffic	Traffic director
CNI	Whole cluster	Connects Nodes and Pods	Road network